
LOZO+: Provably Efficient Zeroth-Order Fine-Tuning via Greedy Low-Rank Subspace Selection

Jinjie Fang¹ Chengxun Jin¹ Tianxing Man² Yi Chang^{1 3 4} Bin Gu¹

Abstract

Zeroth-order (ZO) optimization offers a more memory-efficient alternative to first-order methods for fine-tuning large language models (LLMs). Recent ZO methods, exemplified by LOZO, estimate gradients within low-rank subspaces to align with the low-rank structure of LLM gradients. However, these methods rely on randomly generated subspaces of a fixed rank, which provides no guarantee of alignment with the actual dominant subspaces of the gradients; essentially, they remain ZO gradient descent with stochastic subspace sampling. To more effectively exploit the low-rank nature of LLM gradients, we propose **LOZO+**, an efficient **ZO** fine-tuning algorithm for LLMs that incorporates greedy **Low-Rank** subspace selection. Specifically, LOZO+ leverages loss-based feedback to assess alignment between candidate directions and the dominant low-rank gradient subspaces, and employs an adaptive thresholding criterion to retain only directions yielding substantial gradient descent, thereby steering ZO optimization toward more effective convergence. Importantly, we establish a theoretical framework that characterizes the convergence behavior of LOZO+, formally prove its superiority over existing methods. Extensive experiments demonstrate that LOZO+ consistently outperforms existing ZO methods and achieves performance competitive with FO algorithm, while retaining the memory efficiency inherent to ZO optimization.

1. Introduction

Fine-tuning (FT) large language models (LLMs) for task-specific adaptation has become increasingly important (Bakker et al., 2022; Ding et al., 2023; Lv et al., 2024). Parameter-efficient fine-tuning methods mitigate memory consumption by freezing most pre-trained weights and updating only a small subset of parameters (Hu et al., 2022; Dettmers et al., 2023; Wang et al., 2024). However, despite these advances, as model size continues to scale, first-order (FO) fine-tuning methods that rely on backpropagation still incur substantial memory overhead (Han et al., 2024), often requiring up to $6\times$ more memory than inference ($12\times$ for full-parameter fine-tuning), thereby imposing significant computational constraints (Malladi et al., 2023; Zhao et al., 2024; Ling et al., 2024).

To alleviate this memory bottleneck, zeroth-order (ZO) optimization (Ghadimi & Lan, 2013; Chen et al., 2017) offers a more memory-efficient alternative to FO methods for fine-tuning LLMs (Tang et al., 2024; Zhang et al., 2024; Yang et al., 2024). By approximating gradients via finite differences of function values, ZO methods eliminate the need for backpropagation and activation storage required in FO methods (Yu et al., 2025). Prior ZO approaches for LLM fine-tuning adopt the classical ZO-SGD (Liu et al., 2018), ZO-Adam (Chen et al., 2019) and ZO-SVRG (Liu et al., 2018) framework and achieve memory consumption comparable to inference (Malladi et al., 2023; Liu et al., 2024b; Gautam et al., 2024).

Recent ZO methods estimate gradients within low-rank subspaces to align with the structure of LLM gradients. Extensive prior work has demonstrated that FO gradients generated during backpropagation exhibit low-rank structure (Hu et al., 2022; Liu et al., 2024a; Wang et al., 2024). In contrast, standard ZO gradient estimates are derived from perturbation matrices sampled from Gaussian distributions, which are almost surely full-rank (Ghadimi & Lan, 2013; Wang et al., 2018b). Therefore, standard ZO methods face inherent challenges due to the substantial mismatch in matrix rank between estimated ZO gradients and true FO gradients (Malladi et al., 2023; Zhao et al., 2024; Kozak et al., 2023). To mitigate this discrepancy, LOZO (Chen et al., 2024) performs gradient estimation within low-rank sub-

¹School of Artificial Intelligence, Jilin University, China

²College of Computer Science, Inner Mongolia University, China

³International Center of Future Science, Jilin University, China

⁴Engineering Research Center of Knowledge-Driven Human-Machine Intelligence, MOE, China. Correspondence to: Yi Chang <yichang@jlu.edu.cn>, Bin Gu <gubin@jlu.edu.cn>.

spaces to align with the low-rank structure of LLM gradients. By maintaining low-rank updates throughout optimization, these low-rank ZO methods reduce the variance of gradient estimates in expectation (Yu et al., 2025; Sun et al., 2025; Seung et al., 2025).

However, existing low-rank ZO methods rely on randomly generated fixed-rank subspaces for gradient estimation (Yu et al., 2025; Nozawa et al., 2025; Sun et al., 2025). Typically, at each iteration, a fixed-rank subspace is constructed by randomly sampling matrices from a standard normal distribution, within which ZO gradient estimation is performed via finite-difference evaluations (Chen et al., 2024). Although constraining updates to low-rank subspaces improves alignment with the low-rank structure of LLM gradients, the stochastic nature of subspace sampling provides no guarantee of alignment with the dominant gradient subspace at each iteration. Consequently, these methods can be viewed as ZO gradient descent with stochastic subspace sampling, which limits their ability to consistently exploit effective descent directions and results in suboptimal performance.

To address this issue, we propose **LOZO+**, an efficient **ZO** fine-tuning algorithm that incorporates **greedy Low-Rank** subspace selection to more effectively exploit the low-rank structure of gradients in LLMs. In contrast to existing approaches that rely on stochastic subspace sampling, LOZO+ leverages loss-based feedback to assess the alignment between candidate directions and the dominant low-rank gradient subspace. It further employs an adaptive thresholding criterion to selectively retain directions that yield sufficient descent, enabling progressive refinement of the low-rank subspace and steering ZO optimization toward improved convergence (Figure 1). Our contributions can be summarized as follows:

- Unlike prior low-rank ZO methods such as LOZO, which rely on stochastic subspace sampling, LOZO+ provides a loss feedback mechanism to capture descent-effective low-rank directions.
- We establish a theoretical framework showing that greedy low-rank subspace selection yields strictly improved convergence conditions compared to random subspace sampling, thereby enabling more efficient convergence of LOZO+.
- We conduct extensive experiments across multiple model scales (OPT-13B, Qwen2.5-7B and 32B) and downstream tasks, showing that LOZO+ consistently outperforms existing ZO methods and achieves performance competitive with FO algorithm.

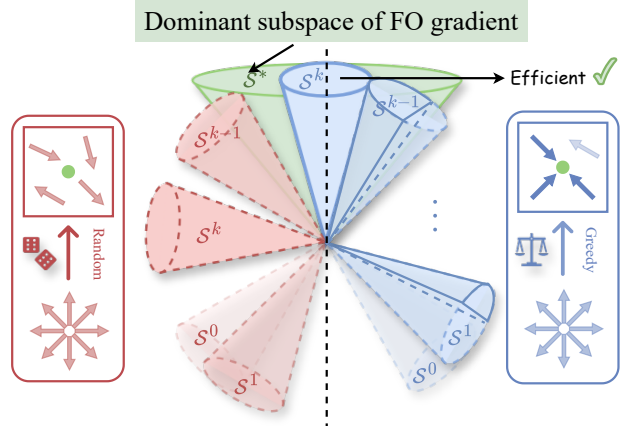


Figure 1. Randomly generated subspace *vs.* greedy subspace selection. The cone represents the subspace, and the solid lines of the cone represent the portion retained by the greedy selection.

2. Related Work

Memory-efficient Fine-tuning LLMs have been widely adopted across diverse downstream applications (Zhang et al., 2022; Touvron et al., 2023; Bai et al., 2023). However, as model scale continues to grow, the increasing demand for task-specific fine-tuning conflicts with the substantial computational and memory overhead of gradient-based optimization, particularly in resource-constrained settings (Hu et al., 2022; Zeng et al., 2024; Mai et al., 2024). To mitigate this challenge, a range of memory-efficient fine-tuning techniques has been proposed to reduce the cost while maintaining competitive performance (Dettmers et al., 2023; Liu et al., 2024a; Qin et al., 2024). Among these methods, ZO optimization has emerged as a compelling alternative. By estimating gradients solely through function evaluations (Ghadimi & Lan, 2013; Nesterov & Spokoiny, 2017), ZO methods eliminate the need for activation storage, enabling efficient optimization when gradient computation is prohibitively memory-intensive (Malladi et al., 2023; Liu et al., 2024b; Tang et al., 2024). MeZO (Malladi et al., 2023) adopts the classical ZO-SGD framework for LLM fine-tuning and achieves memory consumption comparable to inference by eliminating backpropagation overhead. Building upon MeZO, Sparse-MeZO (Liu et al., 2024b) introduces sparse masking to selectively update a subset of parameters, significantly improving performance by reducing the effective optimization dimension. HiZOO (Zhao et al., 2024) further enhances ZO-based fine-tuning by incorporating diagonal Hessian information, addressing convergence instability caused by the highly non-uniform curvature.

Low-rank Zeroth-order Optimization FO gradients produced during LLM fine-tuning exhibit pronounced low-rank structure (Hu et al., 2022; Dettmers et al., 2023; Wang et al., 2024). In contrast, standard ZO fine-tuning methods, exem-

plified by MeZO, estimate gradients from full-rank perturbation matrices sampled from Gaussian distributions (Malladi et al., 2023; Peng et al., 2025; Tan et al., 2025). This mismatch in matrix rank between estimated ZO gradients and the underlying FO gradients can lead to substantial differences in optimization behavior, potentially contributing to the performance gap observed between ZO and FO methods (Chen et al., 2024). To address this issue, recent studies have proposed low-rank ZO optimization algorithms that restrict gradient estimation to low-rank subspaces, thereby better aligning ZO updates with the intrinsic low-rank structure of LLM gradients and improving optimization efficiency and performance (Yu et al., 2025). LOZO (Chen et al., 2024) is a low-rank ZO algorithm designed to capture the intrinsic low-rank structure of gradients observed during LLM fine-tuning, thereby improving optimization efficiency. SubZero (Yu et al., 2025) introduces a randomized subspace ZO optimization framework that applies low-rank perturbations in a layer-wise manner to enable fine-tuning of LLMs.

3. Preliminaries

3.1. Zeroth-order Optimization for LLM Fine-tuning

Formally, we consider the optimization problem:

$$\min_{\mathbf{W}} \mathcal{F}(\mathbf{W}) := \mathbb{E}_{\xi} [\mathcal{F}(\mathbf{W}; \xi)], \quad (1)$$

where ξ denotes the randomness induced by data sampling, and $\mathbf{W}$ denotes the set of trainable parameters. In the context of LLMs fine-tuning, $\mathbf{W} = \{W_{\ell}\}_{\ell=0}^{M-1}$ typically refers to the collection of weight matrices across all M layers, where the index ℓ denotes the ℓ -th layer of the model and each weight matrix satisfies $W_{\ell} \in \mathbb{R}^{m_{\ell} \times n_{\ell}}$.

ZO optimization offers an appealing alternative by relying solely on function evaluations, thereby enabling an efficient and practical approach to optimizing LLMs (Malladi et al., 2023; Chen et al., 2024; Zhao et al., 2024). A widely adopted gradient estimator in this setting is the Simultaneous Perturbation Stochastic Approximation (SPSA) (Ghadimi & Lan, 2013; Chen et al., 2017), which perturbs all coordinates simultaneously using a random vector $\mathbf{Z} \sim \mathcal{N}(0, \mathbf{I}^{m \times n})$:

$$\hat{\nabla} \mathcal{F}(\mathbf{W}; \xi) := \frac{\mathcal{F}(\mathbf{W} + \epsilon \mathbf{Z}; \xi) - \mathcal{F}(\mathbf{W} - \epsilon \mathbf{Z}; \xi)}{2\epsilon} \cdot \mathbf{Z}, \quad (2)$$

which requires only two forward passes and ϵ denotes the perturbation size.

Using the ZO gradient estimator $\hat{\nabla} \mathcal{F}(\mathbf{W}; \xi)$, the ZO stochastic gradient descent (ZO-SGD) proceeds via:

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \cdot \hat{\nabla} \mathcal{F}(\mathbf{W}_t; \xi), \quad (3)$$

where η denotes the step size.

Memory-efficient ZO-SGD. The standard implementation of ZO-SGD incurs substantial memory overhead because constructing the randomized gradient estimator requires explicitly storing the perturbation matrix $\mathbf{Z} \in \mathbb{R}^{m \times n}$. For LLMs, $\mathbf{Z}$ can be comparable in size to the full parameter set, leading to memory usage approaching twice that of inference. To mitigate this limitation, a memory-efficient variant of ZO-SGD, termed Memory-efficient ZO-SGD (MeZO) (Malladi et al., 2023), was proposed to avoid materializing $\mathbf{Z}$. Rather than storing $\mathbf{Z}$ explicitly, MeZO retains only the random seed used to generate the perturbation. When $\mathbf{Z}$ is needed, the random number generator is regenerate with the saved seed.

3.2. Random Low-rank Zeroth-order Optimization

Recent studies show that gradients in LLMs exhibit pronounced low-rank structure, suggesting that effective optimization often occurs within low-dimensional subspaces (Wang et al., 2024; Liu et al., 2024a). However, standard ZO gradient estimators such as SPSA (Ghadimi & Lan, 2013) do not leverage this property. In particular, SPSA perturb parameters using dense Gaussian noise, producing full-rank gradient estimates that fail to capture the intrinsic low-rank structure of LLM gradients. To address this mismatch, a low-rank ZO gradient estimator has been proposed to explicitly preserve low-rank structure during estimation (Chen et al., 2024). Specifically, it samples two low-rank factor matrices $U \in \mathbb{R}^{m \times r}$ and $V \in \mathbb{R}^{n \times r}$ with $r \ll \min\{m, n\}$. The resulting estimator is defined as:

$$\begin{aligned} & \tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi) \\ &:= \frac{\mathcal{F}(\mathbf{W} + \epsilon UV^{\top}; \xi) - \mathcal{F}(\mathbf{W} - \epsilon UV^{\top}; \xi)}{2\epsilon} \cdot UV^{\top}. \end{aligned} \quad (4)$$

4. Methods

In this section, we reveal that random low-rank ZO optimization is fundamentally limited by the weak alignment between randomly sampled subspaces and the dominant FO gradient subspace. To address this limitation, LOZO+ incorporates greedy low-rank subspace selection that refines estimated gradient subspace toward better-aligned (Figure 2).

4.1. From Random Subspaces to Greedy Selection

Why Randomly Sampled Subspaces Are Insufficient.

Let $\nabla \mathcal{F}(\mathbf{W}_t)$ denote the true FO gradient at t -th iteration. Because the gradient exhibits a low-rank structure, we assume the actual dominant subspace is $\mathcal{S}_t^*$ with $\text{Rank}(\mathcal{S}_t^*) \ll \min\{m, n\}$. By performing a Singular Value Decomposition (SVD), we have:

$$\nabla \mathcal{F}(\mathbf{W}_t) = U_t^* \Sigma(V_t^*)^{\top}. \quad (5)$$

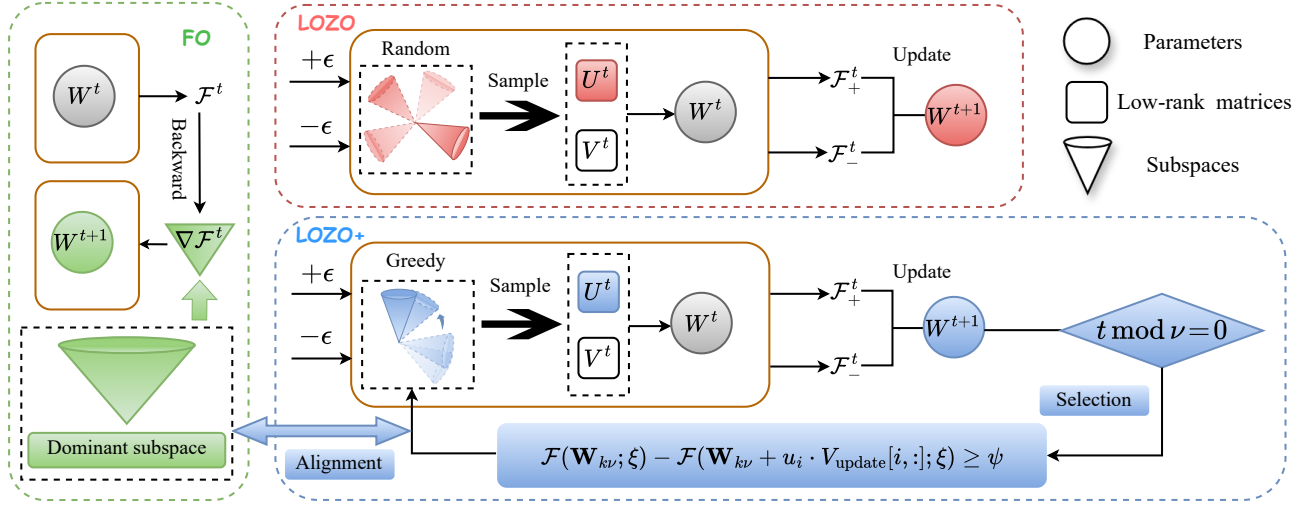


Figure 2. Comparison of the frameworks for LOZO and LOZO+.

We mathematically define $\mathcal{S}_t^*$ as the subspace spanned by the top- r left singular vectors:

$$\mathcal{S}_t^* \triangleq \text{span}(U_t^*[:, 1:r]) \quad (6)$$

In LOZO, the ZO gradient estimator is constructed within a randomly sampled rank- r subspace:

$$\mathcal{S}_t \triangleq \{\text{span}(U) \mid U \sim \mathcal{N}(0, \mathbf{I}^{m \times r})\}, \quad (7)$$

yielding an updated direction:

$$\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) = \mathbf{P}_{\mathcal{S}_t}(\nabla \mathcal{F}(\mathbf{W}_t)) + \phi, \quad (8)$$

where $\mathbf{P}_{\mathcal{S}_t}$ denotes the projection onto $\mathcal{S}_t$, and ϕ denotes the estimation noise arising from finite-difference sampling and stochastic data sampling.

When the subspace is obtained by randomly sampling, there is no guarantee that $\mathcal{S}_t$ aligns with the dominant gradient subspace $\mathcal{S}_t^*$. In particular, the projected gradient magnitude can be interpreted as being governed by both the gradient norm and the alignment between $\mathcal{S}_t$ and $\mathcal{S}_t^*$:

$$\|\mathbf{P}_{\mathcal{S}_t}(\nabla \mathcal{F}(\mathbf{W}_t))\|_F \propto \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F \cos \angle(\mathcal{S}_t, \mathcal{S}_t^*), \quad (9)$$

which can be arbitrarily small even if $r \approx \text{Rank}(\mathcal{S}_t^*)$. In particular, the subspace cosine term $\cos \angle(\mathcal{S}_t, \mathcal{S}_t^*)$ here should be understood in a generalized sense: it does not refer to the cosine similarity of a single pair of gradients, but rather aggregates the angular alignment over all directions spanned by the two subspaces.

Under the L -smoothness assumption and Eq.(8), the de-

crease for objective function per iteration satisfies:

$$\begin{aligned} & \mathcal{F}(\mathbf{W}_{t+1}) - \mathcal{F}(\mathbf{W}_t) \\ & \leq -\eta \cdot \mathbb{E}_{\xi, V} \left[\left\langle \nabla \mathcal{F}(\mathbf{W}_t), \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) \right\rangle \right] \\ & \quad + \frac{L\eta^2}{2} \cdot \mathbb{E}_{\xi, V} \left[\left\| \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) \right\|_F^2 \right] \\ & = -\mathcal{O} \left(\|\mathbf{P}_{\mathcal{S}_t}(\nabla \mathcal{F}(\mathbf{W}_t))\|_F^2 \right) + \mathcal{O}(\phi), \end{aligned} \quad (10)$$

is severely limited, as the descent term is governed by the alignment between the estimated ZO gradient $\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t)$ and the true FO gradient $\nabla \mathcal{F}(\mathbf{W}_t)$.

In other words, although LOZO restricts gradient estimation to low-rank subspaces, it remains fundamentally limited by its reliance on random subspace sampling. The stochastic nature of this process offers no guarantee that the subspace generated at each iteration aligns with the dominant low-rank gradient subspace, thereby hindering the consistent exploitation of effective descent directions.

Toward Greedy Low-rank Subspace. A natural strategy is to increase the squared norm of the projected gradient, $\|\mathbf{P}_{\mathcal{S}_t}(\nabla \mathcal{F}(\mathbf{W}_t))\|_F^2$ in Eq.(10), thereby improving the expected per-iteration decrease of $\mathcal{F}(\mathbf{W}_t)$.

Consequently, without explicitly accessing or manipulating FO gradients, the goal can be recast as improving the alignment between the sampled subspace $\mathcal{S}_t$ and the dominant gradient subspace $\mathcal{S}_t^*$:

$$\rho_t = \cos \angle(\mathcal{S}_t, \mathcal{S}_t^*). \quad (11)$$

4.2. Greedy Low-rank Subspace Selection

The proposed greedy selection mechanism avoids blindly resampling low-rank directions. Instead, it explicitly evalu-

Algorithm 1 LayerwisePerturbation($\mathbf{W}, \epsilon, U, V$)

Require: Weight $\mathbf{W}$, scalar ϵ , matrices U and V .

- 1: **for** $l = 0 \dots M - 1$ **do**
 - 2: $W_l \leftarrow W_l + \epsilon \cdot U_l V_l^\top$
 - 3: **end for**
 - 4: **return** $\mathbf{W}$
-

Algorithm 2 ColumnwiseLoss($\mathbf{W}, \epsilon, U, V_{\text{original}}$)

Require: Weight $\mathbf{W}$, scalar ϵ , matrices U and V_{original} .

- 1: $V_{\text{update}} = (U^{-1} \mathbf{W} - V_{\text{original}})$
 - 2: **for** $i = 0 \dots r - 1$ **do**
 - 3: $\mathcal{F}_{\text{update}}[i] \leftarrow \mathcal{F}(\mathbf{W} - \sum_{j \neq i} u_j \cdot V_{\text{update}}[:, j]; \xi)$
 - 4: **end for**
 - 5: **return** $\mathcal{F}_{\text{update}}$
-

ates the contribution of each direction by loss reduction and selectively retains only those that induce substantial descent. The overall procedure is summarized as follows:

Constructing Greedy Function: Since the dominant gradient subspace $\mathcal{S}_t^*$ is unknown, we adopt an approximate approach to maximize the alignment metric ρ_t . Specifically, we formulate the following surrogate optimization subproblem: for each greedy interval k with iterations $t \in \{k\nu, \dots, (k+1)\nu - 1\}$, we treat the weight matrix at the beginning of the interval $\mathbf{W}_{k\nu}$ as the optimization starting point and the weight matrix at the end of the interval $\mathbf{W}_{(k+1)\nu-1}$ as an approximate endpoint. Then $V_{\text{update}} = U^{-1} (\mathbf{W}_{(k+1)\nu-1} - \mathbf{W}_{k\nu}) \in \mathbb{R}^{r \times n}$ can then be viewed as the coefficient matrix implied by the interval's update. Based on the above, we seek the optimal subspace for this surrogate subproblem, thereby obtaining the approximately optimal subspace for target objective.

We then begin with the orthogonal basis $U = \{u_i\}_{i=0}^{r-1} \in \mathbb{R}^{m \times r}$ and coefficient matrix V_{update} . The relationship between $\mathbf{W}_{k\nu}$ and $\mathbf{W}_{(k+1)\nu-1}$ is given as follows:

$$\mathbf{W}_{(k+1)\nu-1} = \mathbf{W}_{k\nu} + \sum_{i=0}^{r-1} u_i \cdot V_{\text{update}}[i, :]. \quad (12)$$

Based on the above, we construct a greedy criterion for each interval as following:

$$\mathcal{F}(\mathbf{W}_{k\nu}; \xi) - \mathcal{F}(\mathbf{W}_{k\nu} + u_i \cdot V_{\text{update}}[i, :]; \xi), \quad (13)$$

which aims to identify the directions in U that yields the fast loss reduction.

Updating ZO Subspace: We introduce an adaptive threshold criterion (with ψ denoting the threshold) to selectively retain directions that induce sufficient descent, then the re-

tained basis direction u_i satisfy following condition:

$$\mathcal{F}(\mathbf{W}_{k\nu}; \xi) - \mathcal{F}(\mathbf{W}_{k\nu} + u_i \cdot V_{\text{update}}[i, :]; \xi) \geq \psi, \quad (14)$$

where $i \in p_{\text{index}}$ and p_{index} is the corresponding index of the p retained directions. As convergence typically progresses rapidly in the early stages and slows over time, the threshold values are often scheduled to decrease gradually with the number of iterations. This strategy maintains efficient progress while accommodating the diminishing rate of improvement in the later stages of convergence.

Then, the retained part of U can be expressed as:

$$U_{\text{keep}} = U[:, p_{\text{index}}]. \quad (15)$$

To replace the discarded components of U , we construct a random orthogonal matrix U_{new} whose rows are orthogonal to those of U_{keep} . We then concatenate U_{keep} and U_{new} to obtain the updated basis:

$$U = [U_{\text{keep}}, U_{\text{new}}]. \quad (16)$$

Algorithm 3 LOZO+

Require: Weight $\mathbf{W}$, loss function $\mathcal{F}$, samples ξ , learning rate η , perturbation size ϵ , step budget T , number of layers M , interval ν , rank r , threshold ψ .

- 1: Initialize orthogonal basis matrix U
 - 2: **for** $t = 0 \dots T - 1$ **do**
 - 3: Sample $V \in \mathbb{R}^{n \times r}$ from standard normal distribution
 - 4: **if** $t \bmod \nu = 0$ **then**
 - 5: $\mathcal{F}_{\text{original}} \leftarrow \mathcal{F}(\mathbf{W}, \xi)$
 - 6: $V_{\text{original}} \leftarrow U^{-1} \mathbf{W}$
 - 7: **end if**
 - 8: $\mathbf{W} \leftarrow \text{LayerwisePerturbation}(\mathbf{W}, \epsilon, U, V)$
 - 9: $\mathcal{F}_+ \leftarrow \mathcal{F}(\mathbf{W}; \xi)$
 - 10: $\mathbf{W} \leftarrow \text{LayerwisePerturbation}(\mathbf{W}, -2\epsilon, U, V)$
 - 11: $\mathcal{F}_- \leftarrow \mathcal{F}(\mathbf{W}; \xi)$
 - 12: $\mathbf{W} \leftarrow \text{LayerwisePerturbation}(\mathbf{W}, \epsilon, U, V)$
 - 13: $\Phi \leftarrow (\mathcal{F}_+ - \mathcal{F}_-)/(2\epsilon)$
 - 14: $\mathbf{W} \leftarrow \mathbf{W} - (\eta/r) \cdot \Phi \cdot UV^\top$
 - 15: **if** $(t+1) \bmod \nu = 0$ **then**
 - 16: $\mathcal{F}_{\text{update}} \leftarrow \text{ColumnwiseLoss}(\mathbf{W}, \epsilon, U, V_{\text{original}})$
 - 17: $U_{\text{keep}} \leftarrow U[:, \{i \mid \{\mathcal{F}_{\text{original}} - \mathcal{F}_{\text{update}}[i]\}_{i=0}^{r-1} \geq \psi\}]$
 - 18: Construct random orthonormal matrix $U_{\text{new}} \in \mathbb{R}^{m \times (r-p)}$ that is orthogonal to each row of U_{keep}
 - 19: $U \leftarrow [U_{\text{keep}}, U_{\text{new}}]$
 - 20: **end if**
 - 21: **end for**
-

4.3. Design of LOZO+ Algorithm

To reduce memory consumption, we store $V_{\text{original}} \in \mathbb{R}^{r \times n}$ instead of $\mathbf{W}_{k\nu} \in \mathbb{R}^{m \times n}$ for Eq.(14):

$$V_{\text{original}} = U^{-1} \mathbf{W}_{k\nu}. \quad (17)$$

We then perform directional contribution analysis to assess each basis direction:

$$\mathbf{W}_{k\nu} + u_i \cdot V_{\text{update}}[i, :] = \mathbf{W}_{(k+1)\nu-1} - \sum_{j \neq i} u_j \cdot V_{\text{update}}[j, :], \quad (18)$$

where V_{update} can be computed as:

$$V_{\text{update}} = U^{-1} \mathbf{W}_{(k+1)\nu-1} - V_{\text{original}}. \quad (19)$$

All algorithmic details are presented in Algorithm 1-3.

5. Convergence Analysis

Before presenting the convergence analysis, we first introduce several assumptions, which are commonly adopted in the literature on ZO and stochastic optimization methods for nonconvex problems (Ghadimi & Lan, 2013; Nesterov & Spokoiny, 2017; Wang et al., 2018b; Chen et al., 2024):

Assumption 5.1. L-Smoothness: For any ξ , the function $\mathcal{F}(\mathbf{W}; \xi)$ is differentiable with respect to $\mathbf{W}$. Furthermore, the gradient $\nabla \mathcal{F}(\mathbf{W}; \xi)$ is uniformly L -Lipschitz continuous, i.e., $\forall \mathbf{W}_1, \mathbf{W}_2$:

$$\|\nabla \mathcal{F}(\mathbf{W}_1; \xi) - \nabla \mathcal{F}(\mathbf{W}_2; \xi)\|_F \leq L \|\mathbf{W}_1 - \mathbf{W}_2\|_F. \quad (20)$$

Assumption 5.2. Bounded Variance: The stochastic gradient $\mathcal{F}(\mathbf{W}; \xi)$ is unbiased and has bounded variance:

$$\mathbb{E}_\xi [\|\nabla \mathcal{F}(\mathbf{W}; \xi) - \nabla \mathcal{F}(\mathbf{W})\|_F^2] \leq \sigma^2. \quad (21)$$

Assumption 5.3. Bounded initial optimality gap: Given an initial point $\mathbf{W}_0$, and $\mathbf{W}^*$ denotes a global minimizer of $\mathcal{F}$, there exists a constant $\Delta < \infty$ that:

$$\mathcal{F}(\mathbf{W}_0) - \mathcal{F}(\mathbf{W}^*) \leq \Delta. \quad (22)$$

Under the above assumptions, the following theorem characterizes the convergence behavior of the proposed LOZO+ algorithm. The detailed proof is deferred to Appendix B.

Theorem 5.4. Under Assumptions 5.1, 5.2 and 5.3, run the LOZO+ algorithm for $t = 0, \dots, T-1$ steps with period ν , and at each step using the rank- r two-point estimator $\tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi)$, where $U^\top U = I_r$ and $V \in \mathbb{R}^{n \times r}$ has i.i.d. $\mathcal{N}(0, 1)$ entries. Let $\rho^2 = \frac{1}{T} \sum_{t=0}^{T-1} \rho_t^2$ represents the overall similarity between the subspace $\mathcal{S}$ and the dominant true FO gradient space $\mathcal{S}^*$ during the entire iteration process, and $\mathbb{P}(t) = \frac{\rho_t^2}{\sum_{j=0}^{T-1} \rho_j^2}$ is a discrete probability distribution over the iteration index t . Suppose the step size satisfies $0 < \eta < \frac{r}{4L(nr+2)}$. Then the following inequality holds:

$$\mathbb{E}_{t \sim \mathbb{P}} [\|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2] \leq \frac{1}{\rho^2} \frac{4r\Delta}{T\eta} + \frac{1}{\rho^2} G, \quad (23)$$

where $G = \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2} + \eta L^3 \epsilon^2 n(nr+2)(nr+4) + \frac{4\eta L(nr+2)\sigma^2}{r}$.

Remark 5.5 (The role of similarity ρ and feasible greedy threshold ψ). Since the convergence bound in Theorem 5.4 scales as $\frac{1}{\rho^2}$, a larger ρ directly yields a tighter convergence guarantee. ρ_t quantifies the fraction of the true gradient preserved by the estimated subspace. Lemma B.4 further clarifies how the greedy mechanism affects ρ_t . The interval balances quality and reachability:

$$\psi \in [\psi_{l,t}, \psi_{u,t}] \quad (24)$$

The lower bound $\psi \geq \psi_{l,t}$ prevents the greedy mechanism from accepting directions whose contribution is too weak. If ψ is too small, the selected directions may be noisy or poorly aligned with the true gradient, and the resulting subspace may fail to outperform random sampling. The upper bound $\psi \leq \psi_{u,t}$ ensures the reachability of the greedy trigger. If ψ is too large, the selection criterion becomes overly stringent, so no direction can satisfy the greedy condition. In that case, the algorithm degenerates to the purely random subspace case, and ρ_t reduces to the random alignment baseline.

In actual setups, $m \times n$ are exceptionally large ($10^8 \sim 10^9$), r is extremely small (2, 4, 8), η is typically small ($10^{-4} \sim 10^{-3}$), and ν is set between 50 and 200. Therefore, $\psi_{l,t}$ evaluates to an astronomically small value ($10^{-11} \sim 10^{-9}$). In contrast, ψ is set within $[0.01, 0.5]$. Because $\psi \gg \psi_{l,t}$, the lower bound is trivially satisfied; $\psi_{u,t}$ typically falls within a reasonable and macroscopic range ($10^{-2} \sim 1$), so we can easily set ψ to remain safely below this bound.

Remark 5.6 (The guarantee of strict improvement). LOZO+ reverts to the random method when the greedy condition is not met for all iterations. In contrast, if condition $\psi \in [\psi_{l,t}, \psi_{u,t}]$ is satisfied in a single iteration, the greedy mechanism executes a superior update. Since LOZO+ strictly outperforms the random method in triggered steps and matches it in all other steps, the overall global convergence is strictly improved. Practically, LOZO+ operates in this regime far more frequently than “at least once” minimum. In Table 8 of the Appendix A.3, the greedy mechanism is successfully triggered in 30% to 40% of the steps (over 5000 steps) during 20000 iterations. The empirical and theoretical results consistently indicate that LOZO+ strictly improves convergence compared to the random method.

Corollary 5.7. Under the same assumptions of Theorem 5.4, run LOZO+ algorithm with $\eta = T^{-\frac{1}{2}}$ and $\epsilon = T^{-\frac{1}{4}}$, the convergence rate of can be expressed as follows:

$$\mathbb{E}_{t \sim \mathbb{P}} [\|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2] \leq \mathcal{O} \left(\frac{1}{\sqrt{T}} \right). \quad (25)$$

6. Experiments

6.1. Experimental Setup

Datasets and Models. To rigorously evaluate optimization performance, we consider tasks from the GLUE (Wang

et al., 2018a), SuperGLUE (Sarlin et al., 2020), and related benchmarks (e.g., SST-2, RTE, CB, BoolQ, WSC, WiC, MultiRC, COPA, ReCoRD, SQuAD and DROP), which cover a broad range of linguistic phenomena including natural language inference, coreference resolution, common-sense reasoning, and machine reading comprehension. We evaluate all methods using accuracy as the evaluation metric. Each downstream task is restricted to a low-resource setting with only 1,000 labeled training examples. Following standard evaluation protocols, we further construct a validation set of 500 examples for hyperparameter tuning and early stopping, and a test set of 1,000 examples for final performance reporting. We conduct experiments on a diverse set of pretrained language models, including Qwen2.5-7B, Qwen2.5-32B (Bai et al., 2023), OPT-13B, and OPT-66B (Zhang et al., 2022). These models cover different parameter scales, enabling a comprehensive evaluation of the proposed method across representative LLMs.

Baselines. To comprehensively evaluate the effectiveness of our proposed method, we compare it against a diverse set of baselines that represent the current state of the art in LLM optimization. Specifically, we include the memory-efficient ZO baselines: MeZO (Malladi et al., 2023) and MeZO-LoRA (Malladi et al., 2023), the Hessian-informed optimizer HiZOO (Zhao et al., 2024), and the subspace-based methods SubZero (Yu et al., 2025), and low rank ZO method LOZO (Chen et al., 2024). In addition to these ZO-based approaches, we further report results for two widely used non-training baselines: Zero-shot inference (Zero-shot), in-context learning (ICL) (Dong et al., 2024), and two FO baselines: fine-tuning with Adam (FT) (Kingma, 2014) and FT-LoRA (Hu et al., 2022). Detailed descriptions of the baseline methods are provided in the Appendix A.1.

Experiment Configurations. All experiments were conducted on a server equipped with NVIDIA A100 GPUs (80 GB of memory), Intel Xeon CPUs, and 512 GB of RAM. The models were implemented in PyTorch 2.1 and with CUDA 11.8 support. For fair comparison, all baseline methods were re-implemented or run using their officially released code under the same environment and hyperparameter settings whenever possible. Moreover, memory usage across tasks is reported in Tables 6 and 7 in Appendix A.2. For reproducibility, all hyperparameter configurations are summarized in Table 13 in Appendix A.9. Our code is available at <https://github.com/fangjj2000/LOZO-#>.

6.2. Main Results

Performance Comparison Across Models and Tasks. We evaluate the performance of LOZO+ on large-scale decoder-only language models, including Qwen2.5-7B (Table 1), OPT-13B (Table 2), Qwen2.5-32B (Table 3), and

OPT-66B (Table 4). To comprehensively assess the effectiveness and generality of the proposed greedy optimization strategy, we conduct experiments on a diverse set of tasks, covering classification, multiple-choice benchmarks, and generation tasks. As shown in Tables 1-4, LOZO+ consistently outperforms its counterparts across all model scales, yielding overall performance improvements on a wide range of tasks. Notably, the gains are particularly pronounced on several datasets, including classification benchmarks such as RTE, CB, and WSC, as well as generation tasks like SQuAD, demonstrating the effectiveness of the proposed greedy selection.

Optimization Dynamics over Iterations. Figure 3 illustrates the optimization dynamics of different methods in terms of training loss over optimization iterations. Across both the BoolQ and COPA datasets on Qwen2.5-7B, SST2 and WIC datasets on Qwen2.5-32B, the proposed method consistently exhibits faster convergence compared to MeZO and LOZO. In particular, our method achieves a steeper loss reduction in the early stages of training and maintains a more stable descent throughout optimization, indicating improved efficiency and robustness.

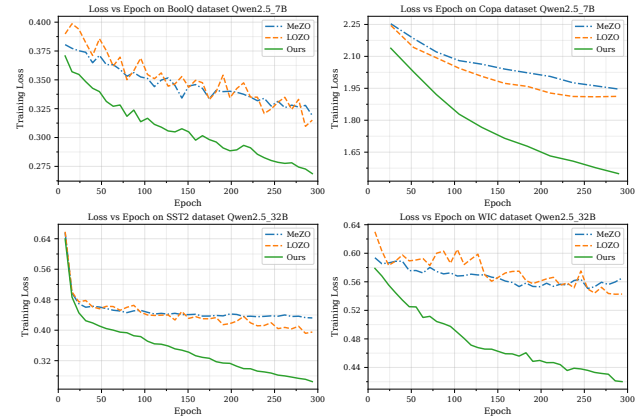


Figure 3. Optimization dynamics of training loss.

6.3. Ablation Study on the Greedy Strategy

To isolate the effect of the proposed greedy subspace selection strategy, we conduct a set of controlled ablation experiments in which the overall optimization framework is held constant and only the subspace selection strategy is varied. Specifically, we compare the following variants:

- Ours-Greedy: which retains basis directions according to the proposed loss-based greedy criterion.
- Ours-NoRefresh: which completely disables subspace selection and refresh.
- Ours-FullReinit: which fully reinitializes the subspace

Table 1. Performance comparison on Qwen2.5-7B across classification, multiple-choice, and generation tasks.

Task Type	Classification							Multiple Choice		Generation		Average
	SST-2	RTE	CB	BoolQ	WSC	WIC	MultiRC	COPA	ReCoRD	SQuAD	DROP	
Zero-shot	63.6	84.5	62.5	81.9	68.3	61.0	77.8	86.0	75.2	85.1	60.8	73.3
ICL	95.3	83.4	91.1	85.4	66.3	64.4	86.7	86.0	76.3	74.8	60.7	79.1
FT	95.0	89.9	96.4	87.6	76.0	75.5	86.6	90.0	80.0	91.5	69.5	85.3
FT-LoRA	79.6	89.5	78.6	80.4	73.1	65.8	87.7	86.0	77.0	89.8	61.9	79.0
MeZO	89.1	88.1	89.3	84.1	69.2	68.8	87.4	89.0	76.9	87.4	60.2	80.9
MeZO-LoRA	78.9	87.4	85.7	82.9	71.2	64.9	86.2	89.0	75.3	81.6	60.7	78.5
SubZero	88.1	88.4	89.3	84.1	72.1	69.3	87.7	90.0	74.6	86.9	59.6	80.9
HiZOO	92.0	88.8	91.1	82.9	65.4	67.1	87.1	86.0	76.7	87.1	60.7	80.4
LOZO	90.6	86.6	96.4	84.3	73.1	70.2	87.9	90.0	76.6	87.5	60.7	82.2
LOZO+	93.1	89.9	96.4	85.5	76.9	70.4	87.4	90.0	78.0	88.4	60.8	83.3

Table 2. Performance comparison on OPT-13B across classification, multiple-choice, and generation tasks.

Task Type	Classification							Multiple Choice		Generation		Average
	SST-2	RTE	CB	BoolQ	WSC	WIC	MultiRC	COPA	ReCoRD	SQuAD	DROP	
Zero-shot	58.8	59.6	46.4	59.0	38.5	55.0	46.9	80.0	81.0	46.2	14.6	53.3
ICL	87.0	62.1	57.1	66.9	39.4	50.5	53.1	87.0	82.3	75.9	29.5	62.8
FT	92.0	70.8	83.9	77.1	63.5	70.1	71.1	79.0	74.1	84.9	31.3	72.5
FT-LoRA	92.4	62.1	60.7	66.9	40.4	58.2	58.0	84.0	81.1	81.3	29.4	65.0
MeZO	91.3	68.2	66.1	68.1	61.5	59.4	59.4	88.0	81.3	81.8	31.3	68.9
MeZO-LoRA	89.6	67.9	67.8	73.5	63.5	60.2	61.3	84.1	81.5	82.1	31.3	69.4
SubZero	92.1	74.0	73.2	71.2	62.5	60.8	61.0	88.0	81.0	84.5	32.0	70.9
HiZOO	92.1	69.3	69.6	63.2	63.5	59.4	61.3	89.0	81.3	80.5	28.2	68.9
LOZO	91.7	70.4	69.6	71.9	63.5	60.8	62.0	89.0	81.3	84.9	30.7	70.5
LOZO+	93.1	74.4	75.0	74.6	63.5	64.9	62.6	89.0	81.7	85.7	31.0	72.3

Table 3. Performance comparison on Qwen2.5-32B across tasks.

Task	SST-2	RTE	BoolQ	WSC	WIC	SQuAD	Average
Zero-shot	61.5	87.0	84.3	80.8	61.8	85.5	76.8
MeZO	87.5	90.3	86.7	81.7	65.7	87.9	83.3
MeZO-LoRA	62.0	87.4	84.8	82.7	64.7	84.7	77.7
SubZero	89.1	90.3	87.7	82.7	68.8	89.6	84.7
LOZO	89.4	91.0	87.0	84.6	69.3	90.5	85.3
LOZO+	93.5	91.3	87.9	84.6	71.8	91.6	86.8

Table 4. Performance comparison on OPT-66B across tasks.

TASK	SST2	RTE	BoolQ	WSC	WIC	SQuAD	Average
Zero-shot	57.5	67.2	66.8	43.3	50.6	48.1	55.6
ICL	89.3	65.3	62.8	52.9	54.9	81.3	67.8
MeZO	92.0	71.5	73.8	64.4	57.8	84.0	73.9
LOZO	92.5	73.6	74.5	63.5	58.7	85.2	74.7
LOZO+	93.6	74.2	76.1	65.2	59.9	86.8	76.0

at fixed intervals without retaining any previously selected directions.

As shown in Table 5, removing the subspace refresh mechanism (Ours-NoRefresh) leads to a clear performance degradation, indicating that subspace updating is essential for effective exploration. Similarly, fully reinitializing the subspace without retaining previously selected directions (Ours-FullReinit) also degrades performance, suggesting that indiscriminate reinitialization disrupts the accumulation of useful directions. Collectively, these results demonstrate that the proposed greedy subspace selection strategy effec-

tively balances exploration and exploitation, and that both components are critical to the observed performance gains.

6.4. Sensitivity Analysis of Greedy Hyperparameters

Sensitivity Analysis of Greedy Threshold. The threshold ψ determines the number of basis directions retained at each greedy interval, thus directly controls the extent of subspace selection and refresh. Accordingly, we vary $\psi \in \{0.2, 0.1, 0.05, 0.01\}$ while fixing r and ν . As reported in Figure 4, an overly large ψ results in retaining too few directions at each update, causing the algorithm to degenerate toward a random subspace sampling method. In contrast,

Table 5. Ablation study on the greedy subspace selection strategy across different models. All variants use the same query budget.

Task	Qwen2.5-7B		OPT-13B		Average
	CB	SQuAD	SST2	WIC	
Ours-Greedy	96.4	88.4	93.1	64.9	85.7
Ours-NoRefresh	89.3	87.8	90.6	58.5	81.7
Ours-FullReinit	94.6	85.7	92.7	59.1	83.0

an excessively small ψ leads to over-retention of directions, weakening the greedy selection mechanism. A properly chosen threshold significantly improves convergence performance. In addition, we report the statistical activation ratios of the greedy mechanism in Table 8 of Appendix A.3.

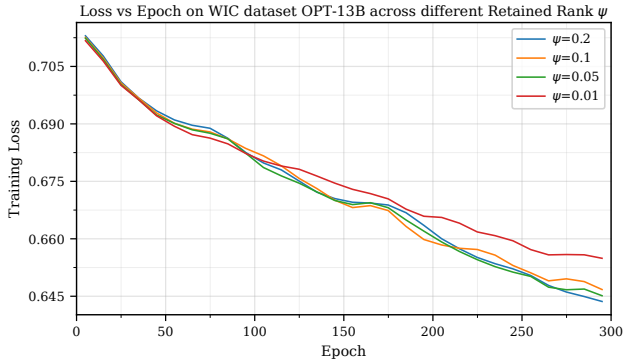


Figure 4. Sensitivity analysis for ψ .

Sensitivity Analysis of Greedy Interval. The greedy interval ν controls how frequently the greedy subspace selection is performed. We evaluate LOZO+ with $\nu \in \{10, 20, 50, 100, 200\}$. As shown in Figure 5, a smaller ν leads to overly frequent subspace updates, which may result in insufficient exploration of subspaces and consequently degrade convergence performance.

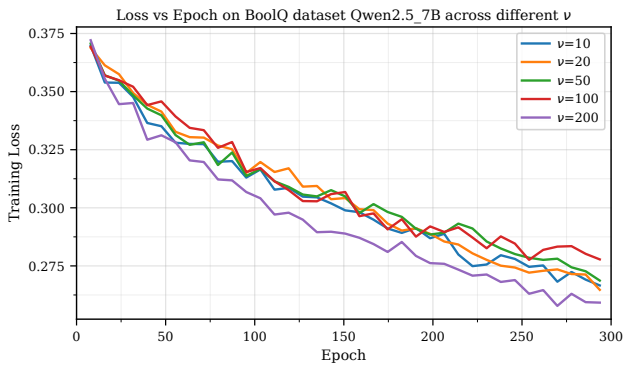


Figure 5. Sensitivity analysis for ν .

Sensitivity Analysis of Subspace Rank. The rank r determines the maximum number of basis directions maintained during optimization. We evaluate LOZO+ with $r \in \{1, 2, 4, 8, 16\}$, while keeping the ψ and ν fixed. As shown in Figure 6, both excessively large and small r can slow training: larger ranks typically lead to higher initial loss and slower early-stage convergence, whereas smaller ranks constrain the subspace dimension.

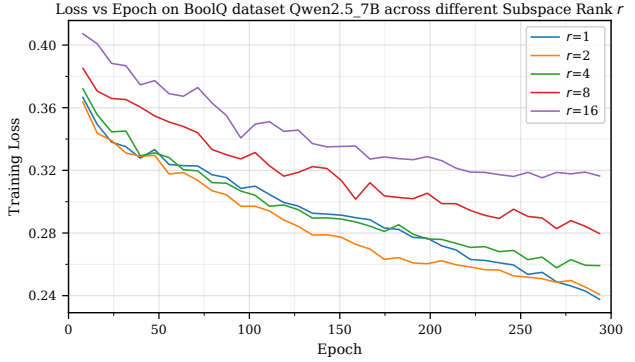


Figure 6. Sensitivity analysis for rank r .

7. Conclusion

In this paper, we propose LOZO+, a greedy low-rank ZO optimization method for memory-efficient fine-tuning of LLMs. Motivated by the pronounced low-rank structure of LLM gradients, LOZO+ introduces a principled mechanism to capture descent-effective low-rank directions. Specifically, LOZO+ leverages loss-based feedback to evaluate the contribution of candidate directions and employs an adaptive thresholding criterion to selectively retain directions that yield sufficient descent. From a theoretical standpoint, our analysis shows that compared to random subspace sampling, LOZO+ admits improved convergence conditions. Empirically, extensive experiments across multiple model scales and downstream tasks demonstrate that LOZO+ consistently outperforms existing ZO methods and achieves performance competitive with FO algorithm, while maintaining inference-level memory efficiency.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Implementation and Experimental Details

A.1. Baseline Descriptions

Detailed descriptions of all baselines are provided below:

- **Zero-shot** evaluates the pretrained model directly without any task-specific adaptation.
- **In-Context Learning** (Dong et al., 2024) performs inference by prepending a small number of task-specific demonstrations to the input prompt, without updating model parameters.
- **Fine-tuning with Adam** (Kingma, 2014) represents a FO optimization baseline, where all model parameters are updated using the Adam optimizer under the same training data budget.
- **MeZO** (Malladi et al., 2023) is a memory-efficient ZO optimization method for fine-tuning large language models without backpropagation. It estimates gradients using symmetric random perturbations and requires only two forward passes per update step. By performing in-place parameter perturbations and avoiding the storage of intermediate activations and backward graphs, MeZO achieves a memory footprint comparable to inference.
- **MeZO-LoRA** (Malladi et al., 2023) extends MeZO by restricting ZO optimization to low-rank adapter parameters introduced by LoRA, while keeping the pretrained backbone frozen. By reducing the effective optimization dimensionality, MeZO-LoRA significantly improves the stability and efficiency of gradient estimation.
- **HiZOO** (Zhao et al., 2024) is a Hessian-informed ZO optimization method designed to improve the stability and efficiency of fine-tuning large language models under memory constraints. By incorporating diagonal Hessian information, HiZOO adaptively scales random perturbations and parameter updates to account for heterogeneous curvature across different parameter dimensions. This Hessian-aware preconditioning alleviates the instability and slow convergence commonly observed in standard ZO methods such as MeZO.
- **SubZero** (Yu et al., 2025) is a random subspace ZO optimization method designed to reduce the variance and improve the efficiency of ZO fine-tuning for large language models. Instead of performing random perturbations in the full parameter space, SubZero restricts gradient estimation to a low-dimensional random subspace, significantly reducing the effective perturbation dimension.
- **LOZO** (Chen et al., 2024) is a low-rank ZO optimization method designed to improve the efficiency of memory-constrained fine-tuning for large language models. By restricting ZO gradient estimation to a low-rank subspace, LOZO captures the intrinsic low-rank structure commonly observed in LLM gradients, thereby significantly reducing estimation variance.

A.2. Memory Consumption

As shown in Tables 6 and 7, LOZO+ achieves a memory footprint comparable to existing ZO methods designed for memory efficiency (HiZOO, SubZero, and LOZO) across all tasks on Qwen2.5-7B and OPT-13B, while remaining substantially more memory-efficient than FT.

Table 6. GPU memory consumption (GB) of different optimization methods on Qwen2.5-7B.

Task	SST2	RTE	CB	BoolQ	WSC	WiC	MultiRC	COPA	ReCoRD	SQuAD	DROP	Average
FT	75.7	109.1	110.1	131.3	79.8	77.4	102.1	74.4	101.0	134.4	108.8	100.4
FT-LoRA	30.6	68.5	58.0	74.3	33.1	29.2	48.6	19.5	78.8	91	98.7	57.3
MeZO	16.4	22.4	23.7	37.4	17.3	17.9	56.9	15.6	26.9	34.0	53.1	29.2
MeZO-LoRA	17.4	22.2	23.6	37.5	16.8	18.0	56.7	15.4	27.1	33.9	54.5	29.4
HiZOO	40.0	46.6	50.0	54.6	41.3	40.0	77.9	39.6	46.9	52.2	69.2	50.8
SubZero	21.4	27.1	25.6	59.8	21.4	21.4	59.4	21.3	36.5	54.9	78.7	38.9
LOZO	21.0	25.3	30.0	46.5	21.0	21.0	59.5	20.9	31.2	46.5	77.4	36.4
LOZO+	25.7	23.6	29.7	36.9	25.0	24.8	57.5	23.4	26.1	35.0	56.0	33.1

Table 7. GPU memory consumption (GB) of different optimization methods on OPT-13B.

Task	SST2	RTE	CB	BoolQ	WSC	WiC	MultiRC	COPA	ReCoRD	SQuAD	DROP	Average
FT	101.9	138.3	145.8	174.9	112.3	108.4	172.6	100.1	110.4	137.4	172.0	134.0
FT-LoRA	38.4	78.6	109.7	142.5	40.9	41.9	107.6	30.5	41.6	175.7	109.9	83.4
MeZO	26.8	34.1	40.1	44.5	27.4	27.2	52.6	25.5	33.8	41.0	74.5	38.9
Mezo-lora	27.2	34.2	40.4	45.0	27.3	27.5	52.9	25.2	33.6	41.2	74.9	39.0
HiZOO	57.2	66.0	72.7	77.0	59.0	58.1	104.1	57.1	64.5	82.9	135.6	75.8
SubZero	29.1	39.5	48.6	45.3	28.4	28.2	63.3	27.4	36.4	48.3	73.7	42.6
LOZO	26.9	35.9	41.1	43.8	28.6	28.0	51.9	26.9	34.8	47.2	76.1	40.1
LOZO+	28.4	36.8	41.8	43.8	30.5	29.8	51.6	28.9	34.9	43.8	75.9	40.6

A.3. Statistics of Greedy Subspace Selection

In Table 8, we report the activation ratio of the greedy mechanism in LOZO+ across all datasets on the Qwen2.5-7B, OPT-13B, and Qwen2.5-32B models. As shown in Table 8, the activation ratios of the greedy thresholding mechanism remain consistently moderate across tasks for both Qwen2.5-7B, OPT-13B, and Qwen2.5-32B.

Table 8. Activation Rate (%) of the Thresholding Mechanism on Qwen2.5-7B, OPT-13B, and Qwen2.5-32B.

Task	SST2	RTE	CB	BoolQ	WSC	WiC	MultiRC	COPA	ReCoRD	SQuAD	DROP
Qwen2.5-7B	17.0	51.5	42.5	22.2	75.3	41.0	36.5	40.5	39.6	47.6	44.3
OPT-13B	40.0	27.5	51.5	33.0	24.5	22.5	30.0	43.0	45.0	52.3	47.1
Qwen2.5-32B	37.8	34.0	-	38.5	50.8	40.8	-	-	-	33.5	-

A.4. Total Function Evaluations

We have added experiments reporting total function evaluations for LOZO+ vs. baselines in Table 9. The results showed that LOZO+ did not significantly increase total function evaluations.

Table 9. Total function evaluations of different optimization methods.

Task	Function Evaluation
MeZO	40000
HiZOO	60000
SubZero	40000
LOZO	40000
LOZO+	40800 ($r = 4$, $\nu = 100$)

A.5. Equal-budget Experiments

We are conducting a stricter, compute-normalized comparison where LOZO+ is evaluated against MeZO, LOZO, and FT with Adam methods under the same total FLOPs budget in Table 10. The results indicate that LOZO+ achieves optimal performance under an equal budget.

A.6. Wall-clock Experiments

We have conducted additional wall-clock time profiling comparing LOZO+ and LOZO on Qwen2.5-7B in Table 11. The results indicate that the per-iteration wall-clock time of LOZO+ is nearly identical to that of LOZO. Since the total training time for ZO optimization is primarily dominated by the forward passes, the periodic subspace updates in LOZO+ do not significantly impact the overall wall-clock (does not require a significant amount of additional forward propagation) efficiency while providing a substantial boost in convergence quality.

Table 10. Equal-budget Total FLOPs Est profiling on SST-2 with Qwen2.5-7B.

Method	Wall-clock (h)	Total FLOPs Est.	Peak GPU Alloc. (GB)	Eval Acc. (%)
LOZO+	2.48	6.96×10^{17}	25.7	93.5
LOZO	2.50	6.96×10^{17}	21.0	92.4
MEZO	3.12	6.96×10^{17}	22.0	89.7
FT with Adam	5.51	6.96×10^{17}	75.7	94.5

Table 11. Wall-clock of different optimization methods on Qwen2.5-7B.

Method	SST2	RTE	CB	BoolQ	WSC	WiC	MultiRC	COPA	ReCoRD	SQuAD
LOZO	2h15m	8h6m	10h12m	10h14m	3h5m	2h50m	23h47m	1h18m	6h32m	9h33m
LOZO+	2h16m	8h4m	10h11m	10h21m	3h8m	2h46m	23h34m	1h18m	6h47m	9h35m

A.7. Principal Cosine Comparison

We calculated the exact cosine similarity between the greedy basis retained over time and the Singular Value Decomposition components of ground-truth FO, comparing these results with the random bases generated by LOZO in Figure 7. The result reveals that the cosine similarity for LOZO+ increases progressively with iterations and maintains a significant margin over LOZO overall, demonstrating its superior alignment with the optimal descent directions.

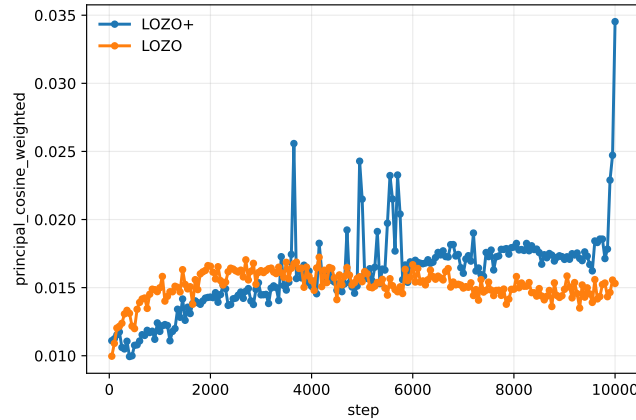


Figure 7. Principal cosine comparison between LOZO+ and LOZO.

A.8. Experiments on Additional Tasks

Table 12. Accuracy on GSM8K task for Qwen2.5-7B.

Methods	Acc
Zero-shot	42.0
FT with Adam	77.0
FT-LoRA	72.0
MeZO	33.0
LOZO	49.0
LOZO+	59.0

A.9. The Hyperparameter Settings

Table 13 summarizes the hyperparameter grids used for LOZO+ and all baselines in our experiments on the Qwen and OPT models. Following the training setup of (Chen et al., 2024; Malladi et al., 2023), we train Qwen and OPT for 20,000 steps. For FT, we use one-tenth as many steps as LOZO+. We save the best checkpoint every one-fifth of the total training steps.

Table 13. The hyperparameter grids used for experiments.

Qwen			OPT		
Experiment	Hyperparameters	Values	Experiment	Hyperparameters	Values
LOZO+	Batch size	16	LOZO+	Batch size	16
	Rank (r)	{2, 4, 8}		Rank (r)	{2, 4, 8}
	Interval (ν)	{50, 100, 200}		Interval (ν)	{50, 100, 200}
	ϵ	{2e-2, 5e-2}		ϵ	{1e-2, 5e-2}
	Learning rate	{1e-3, 5e-4, 1e-4}		Learning rate	{1e-3, 5e-4, 1e-4}
	ψ	{0.1, 0.5}		ψ	0.1
LOZO	Batch size	16	LOZO	Batch size	16
	Rank (r)	{4, 8}		Rank (r)	{2, 4, 8}
	Interval (ν)	{50, 100}		Interval (ν)	{50, 100, 200}
	ϵ	1e-3		ϵ	{1e-3, 1e-4}
	Learning rate	{1e-6, 1e-7}		Learning rate	{1e-6, 1e-7}
HiZOO	Batch size	16	HiZOO	Batch size	16
	ϵ	1e-3		ϵ	1e-3
	Learning rate	1e-6		Learning rate	1e-6
	Hessian smooth type	1e-8		Hessian smooth type	1e-8
SubZero	Batch size	16	SubZero	Batch size	16
	Gauss Rank	24		Gauss Rank	24
	ϵ	1e-3		ϵ	1e-3
	Learning rate	{1e-6, 1e-7}		Learning rate	{1e-6, 1e-7}
	update interval	1000		update interval	1000
	perturbation	two side		perturbation	two side
MeZO	Batch size	16	MeZO	Batch size	16
	ϵ	1e-3		ϵ	1e-3
	Learning rate	{1e-7, 5e-7}		Learning rate	{1e-7, 1e-6, 5e-7}
MeZO-LoRA	Batch size	16	MeZO-LoRA	Batch size	16
	(r, α)	(4, 16)		(r, α)	(8, 16)
	ϵ	1e-2		ϵ	1e-3
	Learning rate	1e-5		Learning rate	{1e-5, 5e-5, 1e-4}
FT	Batch size	8	FT	Batch size	8
	Learning rate	1e-5		Learning rate	{1e-5, 5e-5, 8e-5}
FT-LoRA	Batch size	8	FT-LoRA	Batch size	8
	Learning rate	{1e-5}		Learning rate	{1e-4, 1e-5}
	(r, α)	(8, 16)		(r, α)	(8, 16)

B. Convergence analysis

Lemma B.1. Let $\mathcal{F}(\mathbf{W}; \xi)$ be differentiable in $\mathbf{W} \in \mathbb{R}^{m \times n}$ and L -smooth w.r.t. the Frobenius norm, then:

$$\mathbb{E}_V \left\| \tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi) - \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top \right\|_F^2 \leq \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{4}. \quad (26)$$

Proof. We use Assumption 5.1, which yields

$$\left\| \mathcal{F}(\mathbf{W} + \epsilon UV^\top; \xi) - \mathcal{F}(\mathbf{W}; \xi) - \epsilon \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle \right\|_F \leq \frac{L\epsilon^2}{2} \|UV^\top\|_F^2, \quad (27)$$

$$\left\| \mathcal{F}(\mathbf{W} - \epsilon UV^\top; \xi) - \mathcal{F}(\mathbf{W}; \xi) + \epsilon \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle \right\|_F \leq \frac{L\epsilon^2}{2} \|UV^\top\|_F^2. \quad (28)$$

Combining the above two inequalities, we obtain:

$$\begin{aligned} & \left\| \frac{\mathcal{F}(\mathbf{W} + \epsilon UV^\top; \xi) - \mathcal{F}(\mathbf{W} - \epsilon UV^\top; \xi)}{2\epsilon} - \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle \right\|_F \\ &= \frac{1}{2\epsilon} \left\| \mathcal{F}(\mathbf{W} + \epsilon UV^\top; \xi) - \mathcal{F}(\mathbf{W} - \epsilon UV^\top; \xi) - 2\epsilon \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle \right\|_F \\ &\leq \frac{1}{2\epsilon} \left\| \mathcal{F}(\mathbf{W} + \epsilon UV^\top; \xi) - \mathcal{F}(\mathbf{W}; \xi) - \epsilon \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle \right\|_F \\ &\quad + \frac{1}{2\epsilon} \left\| \mathcal{F}(\mathbf{W} - \epsilon UV^\top; \xi) - \mathcal{F}(\mathbf{W}; \xi) + \epsilon \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle \right\|_F \\ &\leq \frac{1}{2\epsilon} L\epsilon^2 \|UV^\top\|_F^2 = \frac{L\epsilon}{2} \|UV^\top\|_F^2. \end{aligned} \quad (29)$$

Multiplying both sides by $\|UV^\top\|_F$ and squaring both sides, and then taking the expectation with respect to V yields:

$$\mathbb{E}_V \left\| \tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi) - \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top \right\|_F^2 \leq \frac{L^2 \epsilon^2}{4} \mathbb{E}_V \|UV^\top\|_F^6. \quad (30)$$

Based on the column orthogonality of U :

$$\|UV^\top\|_F^2 = \text{tr}((UV^\top)^\top (UV^\top)) = \text{tr}(VU^\top UV^\top) = \text{tr}(VV^\top) = \|V\|_F^2. \quad (31)$$

Then, we can get:

$$\mathbb{E}_V \left\| \tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi) - \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top \right\|_F^2 \leq \frac{L^2 \epsilon^2}{4} \mathbb{E}_V \|V\|_F^6. \quad (32)$$

By exploiting the sixth-order moment identity of standard Gaussian variables, we obtain $\mathbb{E}_V \|V\|_F^6 = nr(nr+2)(nr+4)$, which yields:

$$\mathbb{E}_V \left\| \tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi) - \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top \right\|_F^2 \leq \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{4}. \quad (33)$$

□

Lemma B.2. Let $\mathcal{F}(\mathbf{W}; \xi)$ be differentiable in $\mathbf{W} \in \mathbb{R}^{m \times n}$, then the projection of the true FO gradient onto the subspace satisfies the following equation:

$$\mathbb{E}_V [\nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top] UV^\top = UU^\top \nabla \mathcal{F}(\mathbf{W}; \xi). \quad (34)$$

Proof. Let $B := U^\top \nabla \mathcal{F}(\mathbf{W}; \xi) \in \mathbb{R}^{r \times n}$, using the invariance under transposition, then:

$$\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle = \text{tr}(\nabla \mathcal{F}(\mathbf{W}; \xi)^\top UV^\top) = \text{tr}(VU^\top \nabla \mathcal{F}(\mathbf{W}; \xi)) = \text{tr}(VB) = \sum_{a=1}^n \sum_{b=1}^r V_{a,b} B_{b,a}. \quad (35)$$

Then, we multiply both sides by $V_{j,i}$ where $j = 1, \dots, r$ and $i = 1, \dots, n$:

$$\mathbb{E}_V[\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle V_{j,i}] = \sum_{a=1}^n \sum_{b=1}^r B_{b,a} \mathbb{E}_V[V_{a,b} V_{j,i}]. \quad (36)$$

Since the entries of $V_{(\cdot, \cdot)}$ are independent standard Gaussian random variables, their second moments satisfy:

$$\mathbb{E}_V[V_{a,b} V_{j,i}] = \delta_{aj} \delta_{bi}, \quad (37)$$

where $\delta_{(\cdot, \cdot)}$ denotes the Kronecker delta. Therefore:

$$\mathbb{E}_V[\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle V_{j,i}] = B_{i,j}. \quad (38)$$

This shows that:

$$\mathbb{E}_V[\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle V^\top] = B. \quad (39)$$

We can get:

$$\mathbb{E}_V[\nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top] UV^\top = U \mathbb{E}_V[\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle V^\top] = UB = UU^\top \nabla \mathcal{F}(\mathbf{W}; \xi). \quad (40)$$

□

Lemma B.3. For the gradient estimator $\tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi)$, the following bound holds:

$$\mathbb{E}_V \|\tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi)\|_F^2 \leq 2(nr+2) \|UU^\top \nabla \mathcal{F}(\mathbf{W}; \xi)\|_F^2 + \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2}. \quad (41)$$

Proof. The triangle inequality provides the following bound:

$$\mathbb{E}_V \|\tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi)\|_F^2 \leq 2 \mathbb{E}_V \|\tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi) - \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top\|_F^2 + 2 \mathbb{E}_V \|\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top\|_F^2. \quad (42)$$

To bound the first term, we use Lemma B.1, which yields

$$\mathbb{E}_V \|\tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi) - \langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top\|_F^2 \leq \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{4}. \quad (43)$$

Then, we consider the second term. We first rewrite the inner product as a Frobenius inner product with respect to V , using the cyclic property of the trace:

$$\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle = \text{tr}(\nabla \mathcal{F}(\mathbf{W}; \xi)^\top UV^\top) = \text{tr}(V^\top \nabla \mathcal{F}(\mathbf{W}; \xi)^\top U) = \langle V, \nabla \mathcal{F}(\mathbf{W}; \xi)^\top U \rangle. \quad (44)$$

Then, the above expectation can be written as:

$$\begin{aligned} & \mathbb{E}_V \|\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top\|_F^2 \\ &= \mathbb{E}_V \|\langle V, \nabla \mathcal{F}(\mathbf{W}; \xi)^\top U \rangle UV^\top\|_F^2 \\ &= \mathbb{E}_V \|\langle V, \nabla \mathcal{F}(\mathbf{W}; \xi)^\top U \rangle\|_F^2 \|UV^\top\|_F^2 \\ &\stackrel{(1)}{=} \mathbb{E}_V \|\langle V, \nabla \mathcal{F}(\mathbf{W}; \xi)^\top U \rangle\|_F^2 \|V^\top\|_F^2, \end{aligned} \quad (45)$$

where (1) uses the column orthogonality of U .

Since V has i.i.d. standard Gaussian entries, we obtain:

$$\begin{aligned} & \mathbb{E}_V \|\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle UV^\top\|_F^2 \\ &= \mathbb{E}_V [\|\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle\|_F^2 \cdot \|UV^\top\|_F^2] \\ &= \mathbb{E}_V \|V\|_F^2 \cdot \mathbb{E}_V \|\langle \nabla \mathcal{F}(\mathbf{W}; \xi), UV^\top \rangle\|_F^2 \\ &= \mathbb{E}_V \|V\|_F^4 \cdot \|\langle \nabla \mathcal{F}(\mathbf{W}; \xi), U \rangle\|_F^2 \\ &\stackrel{(1)}{=} (nr+2) \|\nabla \mathcal{F}(\mathbf{W}; \xi)^\top U\|_F^2 \\ &= (nr+2) \|UU^\top \nabla \mathcal{F}(\mathbf{W}; \xi)\|_F^2, \end{aligned} \quad (46)$$

where (1) follows from the fourth-order moment identity of standard Gaussian variables.

Combining these two term Eq.(43) and Eq.(46) to Eq.(42):

$$\mathbb{E}_V \|\tilde{\nabla} \mathcal{F}(\mathbf{W}; \xi)\|_F^2 \leq 2(nr+2) \|UU^\top \nabla \mathcal{F}(\mathbf{W}; \xi)\|_F^2 + \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2}. \quad (47)$$

□

Lemma B.4. *The projection of the true FO gradient onto the estimated gradient subspace satisfies the following:*

$$\|UU^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F = \rho_t \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F \quad (48)$$

Proof. According to $U = [U_{\text{keep}}, U_{\text{new}}]$, we can obtain:

$$\begin{aligned} & \|UU^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F \\ &= \|[U_{\text{keep}}, U_{\text{new}}][U_{\text{keep}}, U_{\text{new}}]^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F \\ &\stackrel{(1)}{=} \underbrace{\|U_{\text{keep}} U_{\text{keep}}^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F}_a + \underbrace{\|U_{\text{new}} U_{\text{new}}^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F}_b, \end{aligned} \quad (49)$$

where (1) uses the orthogonality between U_{keep} and U_{new} .

For term a): Let $i = 0, \dots, p_{\text{keep}} - 1$ direction are retained in U_{keep} . Then, based on the smoothness of $\mathcal{F}(\mathbf{W})$, we can obtain:

$$\mathcal{F}(\mathbf{W}_t + U_{\text{keep}}[:, i] V_{\text{update}}[i, :]) - \mathcal{F}(\mathbf{W}_t) \leq \langle \nabla \mathcal{F}(\mathbf{W}_t), U_{\text{keep}}[:, i] V_{\text{update}}[i, :] \rangle + \frac{L}{2} \|U_{\text{keep}}[:, i] V_{\text{update}}[i, :]\|_F^2. \quad (50)$$

Then, according to the threshold ψ of the greedy mechanism, we can obtain:

$$\langle \nabla \mathcal{F}(\mathbf{W}_t), U_{\text{keep}}[:, i] V_{\text{update}}[i, :] \rangle + \frac{L}{2} \|U_{\text{keep}}[:, i] V_{\text{update}}[i, :]\|_F^2 \geq \psi. \quad (51)$$

Complete the square and then square both sides to get:

$$\left\| U_{\text{keep}}[:, i] V_{\text{update}}[i, :] + \frac{1}{L} \nabla \mathcal{F}(\mathbf{W}_t) \right\|_F^2 \geq \frac{2\psi}{L} + \frac{1}{L^2} \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 \quad (52)$$

$$\|U_{\text{keep}}[:, i] V_{\text{update}}[i, :]\|_F \geq \sqrt{\frac{2\psi}{L} + \frac{1}{L^2} \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2} - \left\| \frac{1}{L} \nabla \mathcal{F}(\mathbf{W}_t) \right\|_F \quad (53)$$

According to the definition of V_{update} and Lemma B.2, we can obtain:

$$\|U_{\text{keep}} U_{\text{keep}}^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F \geq \frac{p}{\eta\nu} \left(\sqrt{\frac{2\psi}{L} + \frac{1}{L^2} \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2} - \left\| \frac{1}{L} \nabla \mathcal{F}(\mathbf{W}_t) \right\|_F \right) \quad (54)$$

For term b):

$$\mathbb{E} [\|U_{\text{new}} U_{\text{new}}^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F] = \frac{r-p}{m-p} (\|\nabla \mathcal{F}(\mathbf{W}_t)\|_F - \|U_{\text{keep}} U_{\text{keep}}^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F) \quad (55)$$

Substituting a) and b) to Eq.(49) get:

$$\|UU^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F \geq \frac{p(m-r)}{\eta\nu(m-p)} \left(\sqrt{\frac{2\psi}{L} + \frac{1}{L^2} \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2} - \left\| \frac{1}{L} \nabla \mathcal{F}(\mathbf{W}_t) \right\|_F \right) + \frac{r-p}{m-p} \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F \quad (56)$$

Upper Bound for ψ : Let $\psi_{u,t}$ represent the maximum contribution that the greedy mechanism can identify. If ψ exceeds this bound, the selection criterion becomes too stringent, resulting in zero retained directions and causing the

algorithm to stall or fail to trigger. Thus, the upper bound ensures the reachability and activation of the greedy mechanism. Based on the (approximate) unbiasedness of stochastic and ZO gradients:

$$\psi \leq -\eta\nu \langle \nabla \mathcal{F}(\mathbf{W}), U_{\text{keep}} U_{\text{keep}}^\top \nabla \mathcal{F}(\mathbf{W}) \rangle + \frac{L\eta^2\nu^2}{2} \|U_{\text{keep}} U_{\text{keep}}^\top \nabla \mathcal{F}(\mathbf{W})\|^2. \quad (57)$$

Based on the orthogonality of U_{keep} and U_{new} and the $0 < \eta < \frac{r}{4L(nr+2)}$ in Theorem 5.4, we can get $\psi_{u,t}$:

$$\psi \leq \psi_{u,t} = \frac{\eta\nu}{p} \left(\frac{\nu}{4} - 1 \right) \|U_{\text{keep}} U_{\text{keep}}^\top \nabla \mathcal{F}(\mathbf{W}_t)\|^2 \quad (58)$$

Lower Bound for ψ : Let $\psi_{l,t}$ be a statistical baseline to ensure that the alignment between the greedy directions and the true gradient strictly outperforms random sampling. When the greedy mechanism is not triggered ($p = 0$), our algorithm reduces to the randomized case with $\rho_{\text{random}} = r/m$. This recovers the standard convergence bound of LOZO, indicating that the gap between the two methods is merely a constant-level factor and does not alter the fundamental convergence rate. The ψ falls below this threshold (or $\psi = 0$), which will lead to low-quality updates that are inferior to random search. Consequently, the lower bound guarantees the superiority of the greedy trigger. When $\rho_{\text{greedy}} \geq \rho_{\text{random}}$ get the lower bound $\psi_{l,t}$:

$$\psi \geq \psi_{l,t} = \frac{\eta\nu}{m} \left(\frac{r\nu}{8m(nr+2)} + 1 \right) \|\nabla \mathcal{F}(\mathbf{W}_t)\|^2 \quad (59)$$

The interval $\Psi_t = [\psi_{l,t}, \psi_{u,t}]$ defines a feasible search envelope that enables the greedy algorithm to consistently select directions that are more closely aligned with the true gradient subspace, thereby maximizing ρ .

Argument for Upper and Lower Bound: In actual setups, $m \times n$ are exceptionally large ($10^8 \sim 10^9$), r is extremely small (2, 4, 8), η is typically small ($10^{-4} \sim 10^{-3}$), and ν is set between 50 and 200. Therefore, $\psi_{l,t}$ evaluates to an astronomically small value ($10^{-11} \sim 10^{-9}$). In contrast, ψ is set within $[0.01, 0.5]$. Because $\psi \gg \psi_{l,t}$, the lower bound is trivially satisfied; $\psi_{u,t}$ typically falls within a reasonable and macroscopic range ($10^{-2} \sim 1$), we can easily set ψ to remain safely below this bound.

The Guarantee of Strict Improvement: LOZO+ reverts to the random method when the greedy condition is not met for all iterations. In contrast, if condition $\psi \in [\psi_{l,t}, \psi_{u,t}]$ is satisfied in a single iteration, the greedy mechanism executes a superior update. Since LOZO+ strictly outperforms the random method in triggered steps and matches it in all other steps, the overall global convergence is strictly improved. Practically, LOZO+ operates in this regime far more frequently than "at least once" minimum. In Appendix Table 8 of the manuscript, the greedy mechanism is successfully triggered in 30% to 40% of the steps (over 5000 steps) during 20000 iterations. The empirical and theoretical results consistently indicate that LOZO+ strictly improves convergence compared to the random method. $\square$

Proof of Theorem 5.4: By Assumption 5.1, the smoothness of function $\mathcal{F}(\mathbf{W}; \xi)$ implies the smoothness of function $\mathcal{F}(\mathbf{W})$, then:

$$\begin{aligned} \mathcal{F}(\mathbf{W}_{t+1}) &\leq \mathcal{F}(\mathbf{W}_t) + \langle \nabla \mathcal{F}(\mathbf{W}_t), \mathbf{W}_{t+1} - \mathbf{W}_t \rangle + \frac{L}{2} \|\mathbf{W}_{t+1} - \mathbf{W}_t\|_F^2 \\ &= \mathcal{F}(\mathbf{W}_t) - \frac{\eta}{r} \langle \nabla \mathcal{F}(\mathbf{W}_t), \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi) \rangle + \frac{L\eta^2}{2r^2} \|\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t)\|_F^2. \end{aligned} \quad (60)$$

For term $\langle \nabla \mathcal{F}(\mathbf{W}_t), \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) \rangle$:

$$\begin{aligned}
 & \langle \nabla \mathcal{F}(\mathbf{W}_t), \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) \rangle \\
 &= \text{tr} \left(\nabla \mathcal{F}(\mathbf{W}_t)^\top \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) \right) \\
 &= \text{tr} \left(\nabla \mathcal{F}(\mathbf{W}_t)^\top U \alpha V^\top \right) \\
 &\stackrel{(1)}{=} \text{tr} \left(\nabla \mathcal{F}(\mathbf{W}_t)^\top U U^\top U \Phi V^\top \right) \\
 &= \text{tr} \left((U U^\top \nabla \mathcal{F}(\mathbf{W}_t))^\top U \Phi V^\top \right) \\
 &= \langle U U^\top \nabla \mathcal{F}(\mathbf{W}_t), \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) \rangle \\
 &\stackrel{(2)}{=} \langle \mathbb{E}_V [\nabla \mathcal{F}(\mathbf{W}_t), U V^\top] U V^\top, \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) \rangle \\
 &= \langle \mathbb{E}_V [\nabla \mathcal{F}(\mathbf{W}_t), U V^\top] U V^\top, \langle \nabla \mathcal{F}(\mathbf{W}_t; \xi_t), U V^\top \rangle U V^\top \rangle \\
 &\quad + \langle \mathbb{E}_V [\nabla \mathcal{F}(\mathbf{W}_t), U V^\top] U V^\top, \tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) - \langle \nabla \mathcal{F}(\mathbf{W}_t; \xi_t), U V^\top \rangle U V^\top \rangle \\
 &\stackrel{(3)}{\geq} \langle \mathbb{E}_V [\nabla \mathcal{F}(\mathbf{W}_t), U V^\top] U V^\top, \langle \nabla \mathcal{F}(\mathbf{W}_t; \xi_t), U V^\top \rangle U V^\top \rangle \\
 &\quad - \frac{1}{2} \|U U^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 - \frac{1}{2} \|\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) - \langle \nabla \mathcal{F}(\mathbf{W}_t; \xi_t), U V^\top \rangle U V^\top\|_F^2,
 \end{aligned} \tag{61}$$

where (1) uses the column orthogonality of U ; (2) uses Lemma B.2; (3) uses inequality $\frac{1}{2}a^2 + \frac{1}{2}b^2 \geq ab$.

Take the expectation of both sides of the Eq.(61) with respect to V, ξ_t :

$$\begin{aligned}
 & \langle \nabla \mathcal{F}(\mathbf{W}_t), \mathbb{E}_{V, \xi_t} [\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t)] \rangle \\
 &\geq \frac{1}{2} \|U U^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 - \frac{1}{2} \mathbb{E}_{V, \xi_t} \|\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) - \langle \nabla \mathcal{F}(\mathbf{W}_t; \xi_t), U V^\top \rangle U V^\top\|_F^2,
 \end{aligned} \tag{62}$$

Then, substitute Eq.(62) into Eq.(60):

$$\begin{aligned}
 & \mathcal{F}(\mathbf{W}_{t+1}) - \mathcal{F}(\mathbf{W}_t) \\
 &\leq -\frac{\eta}{2r} \|U U^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 + \frac{\eta}{2r} \underbrace{\mathbb{E}_{V, \xi_t} \|\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) - \langle \nabla \mathcal{F}(\mathbf{W}_t; \xi_t), U V^\top \rangle U V^\top\|_F^2}_a + \frac{L\eta^2}{2r^2} \underbrace{\mathbb{E}_{V, \xi_t} \|\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t)\|_F^2}_b.
 \end{aligned} \tag{63}$$

For term a), according to Lemma B.1:

$$\mathbb{E}_{V, \xi_t} \|\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t) - \langle \nabla \mathcal{F}(\mathbf{W}_t; \xi_t), U V^\top \rangle U V^\top\|_F^2 \leq \frac{L^2 \epsilon^2 n r (n r + 2)(n r + 4)}{4} \tag{64}$$

For term b), according to Lemma B.3:

$$\begin{aligned}
 & \mathbb{E}_{V, \xi_t} \|\tilde{\nabla} \mathcal{F}(\mathbf{W}_t; \xi_t)\|_F^2 \\
 &\leq 2(n r + 2) \mathbb{E}_{\xi_t} \|U U^\top \nabla \mathcal{F}(\mathbf{W}_t; \xi_t)\|_F^2 + \frac{L^2 \epsilon^2 n r (n r + 2)(n r + 4)}{2} \\
 &= 2(n r + 2) \mathbb{E}_{\xi_t} \|U U^\top \nabla \mathcal{F}(\mathbf{W}) + U U^\top (\nabla \mathcal{F}(\mathbf{W}_t; \xi_t) - \nabla \mathcal{F}(\mathbf{W}))\|_F^2 + \frac{L^2 \epsilon^2 n r (n r + 2)(n r + 4)}{2} \\
 &= 2(n r + 2) \mathbb{E}_{\xi_t} [\|U U^\top \nabla \mathcal{F}(\mathbf{W})\|_F^2 + \|U U^\top (\nabla \mathcal{F}(\mathbf{W}_t; \xi_t) - \nabla \mathcal{F}(\mathbf{W}))\|_F^2] \\
 &\quad + 2(n r + 2) \mathbb{E}_{\xi_t} [2 \langle U U^\top \nabla \mathcal{F}(\mathbf{W}), U U^\top (\nabla \mathcal{F}(\mathbf{W}_t; \xi_t) - \nabla \mathcal{F}(\mathbf{W})) \rangle] + \frac{L^2 \epsilon^2 n r (n r + 2)(n r + 4)}{2} \\
 &\stackrel{(1)}{=} 2(n r + 2) \mathbb{E}_{\xi_t} [\|U U^\top \nabla \mathcal{F}(\mathbf{W})\|_F^2 + \|U U^\top (\nabla \mathcal{F}(\mathbf{W}_t; \xi_t) - \nabla \mathcal{F}(\mathbf{W}))\|_F^2] + \frac{L^2 \epsilon^2 n r (n r + 2)(n r + 4)}{2} \\
 &\stackrel{(2)}{\leq} 2(n r + 2) (\|U U^\top \nabla \mathcal{F}(\mathbf{W})\|_F^2 + \sigma^2) + \frac{L^2 \epsilon^2 n r (n r + 2)(n r + 4)}{2},
 \end{aligned} \tag{65}$$

where (1) uses the unbiasedness of gradient $\nabla \mathcal{F}(\mathbf{W}; \xi)$ in Assumption 5.2; (2) use the bounded variance in Assumption 5.2.

Substituting two terms a) and b), we get:

$$\begin{aligned}
 & \mathcal{F}(\mathbf{W}_{t+1}) - \mathcal{F}(\mathbf{W}_t) \\
 & \leq -\frac{\eta}{2r} \|UU^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 + \frac{\eta}{2r} \left(\frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{4} \right) \\
 & \quad + \frac{L\eta^2}{2r^2} \left(2(nr+2) (\|UU^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 + \sigma^2) + \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2} \right) \\
 & = \left(-\frac{\eta}{2r} + \frac{L\eta^2(nr+2)}{r^2} \right) \|UU^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 + \frac{\eta L^2 \epsilon^2 n(nr+2)(nr+4)}{8} \\
 & \quad + \frac{\eta^2 L^3 \epsilon^2 n(nr+2)(nr+4)}{4r} + \frac{\eta^2 L(nr+2)\sigma^2}{r^2}.
 \end{aligned} \tag{66}$$

Let $-\frac{\eta}{2r} + \frac{L\eta^2(nr+2)}{r^2} < 0$, then $0 < \eta < \frac{r}{2L(nr+2)}$, we can get:

$$\begin{aligned}
 & \left(\frac{\eta}{2r} - \frac{L\eta^2(nr+2)}{r^2} \right) \|UU^\top \nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 \\
 & \leq \mathcal{F}(\mathbf{W}_t) - \mathcal{F}(\mathbf{W}_{t+1}) + \frac{\eta L^2 \epsilon^2 n(nr+2)(nr+4)}{8} + \frac{\eta^2 L^3 \epsilon^2 n(nr+2)(nr+4)}{4r} + \frac{\eta^2 L(nr+2)\sigma^2}{r^2}.
 \end{aligned} \tag{67}$$

By Lemma B.4, we can get:

$$\begin{aligned}
 & \left(\frac{\eta}{2r} - \frac{L\eta^2(nr+2)}{r^2} \right) \rho_t^2 \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 \\
 & \leq \mathcal{F}(\mathbf{W}_t) - \mathcal{F}(\mathbf{W}_{t+1}) + \frac{\eta L^2 \epsilon^2 n(nr+2)(nr+4)}{8} + \frac{\eta^2 L^3 \epsilon^2 n(nr+2)(nr+4)}{4r} + \frac{\eta^2 L(nr+2)\sigma^2}{r^2}.
 \end{aligned} \tag{68}$$

By summing over $t = 0, \dots, T-1$ and divide both sides by T , we further let $0 < \eta < \frac{r}{4L(nr+2)}$, we obtain:

$$\frac{1}{T} \sum_{t=0}^{T-1} \rho_t^2 \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 \leq \frac{4r\Delta}{T\eta} + \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2} + \eta L^3 \epsilon^2 n(nr+2)(nr+4) + \frac{4\eta L(nr+2)\sigma^2}{r}. \tag{69}$$

Let $\rho^2 = \frac{1}{T} \sum_{t=0}^{T-1} \rho_t^2$, and divide both sides by ρ^2 :

$$\begin{aligned}
 & \sum_{t=0}^{T-1} \frac{\rho_t^2}{\sum_{j=0}^{T-1} \rho_j^2} \|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2 \\
 & \leq \frac{1}{\rho^2} \left(\frac{4r\Delta}{T\eta} + \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2} + \eta L^3 \epsilon^2 n(nr+2)(nr+4) + \frac{4\eta L(nr+2)\sigma^2}{r} \right).
 \end{aligned} \tag{70}$$

Consider the left side as a weighted expectation with respect to the iteration index t , we define a discrete probability distribution over the iteration index $t \in \{0, \dots, T-1\}$ as $\mathbb{P}(t) = \frac{\rho_t^2}{\sum_{j=0}^{T-1} \rho_j^2}$, then:

$$\mathbb{E}_{t \sim \mathbb{P}} [\|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2] \leq \frac{1}{\rho^2} \frac{4r\Delta}{T\eta} + \frac{1}{\rho^2} \left(\frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2} + \eta L^3 \epsilon^2 n(nr+2)(nr+4) + \frac{4\eta L(nr+2)\sigma^2}{r} \right) \tag{71}$$

Let $G = \frac{L^2 \epsilon^2 nr(nr+2)(nr+4)}{2} + \eta L^3 \epsilon^2 n(nr+2)(nr+4) + \frac{4\eta L(nr+2)\sigma^2}{r}$, and we can get:

$$\mathbb{E}_{t \sim \mathbb{P}} [\|\nabla \mathcal{F}(\mathbf{W}_t)\|_F^2] \leq \frac{1}{\rho^2} \frac{4r\Delta}{T\eta} + \frac{1}{\rho^2} G \tag{72}$$

□